Triangle Centers, Ceva, Menelaus & Stewart Theorem

Triangle Centers, Ceva, Menelaus & Stewart Theorem Practice Exercises

1 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{1}{2}$ and $\frac{CE}{EA} = \frac{1}{3}$, find $\frac{AF}{FB}$.

a. $\frac{1}{6}$

b. $\frac{3}{2}$

c. 6

d. $\frac{5}{2}$

2 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{2}{3}$ and $\frac{CE}{EA} = \frac{1}{4}$, find $\frac{AF}{FB}$.

a. $\frac{1}{6}$

b. 6

c. $\frac{8}{3}$

d. $\frac{7}{3}$

3 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{3}{4}$ and $\frac{CE}{EA} = \frac{2}{5}$, find $\frac{AF}{FB}$.

a. $\frac{10}{3}$

b. $\frac{3}{10}$

c. $\frac{15}{8}$

d. $\frac{9}{5}$

4 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{1}{4}$ and $\frac{CE}{EA} = \frac{3}{2}$, find $\frac{AF}{FB}$.

a. $\frac{3}{8}$

b. $\frac{1}{6}$

c. $\frac{3}{2}$

d. $\frac{8}{3}$

5 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 2$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{2}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{3}$

b. $\frac{5}{2}$

c. $\frac{2}{5}$

d. $\frac{3}{5}$

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6 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 3$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{3}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$

c. 3

d. $\frac{1}{3} + 1$

7 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 4$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{4}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{7}$

b. $\frac{2}{3}$

c. $\frac{7}{2}$

d. $\frac{3}{14}$

8 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 6$ and $CD = 2$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{5}{3}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{1}{5}$

b. $\frac{20}{3}$

c. $\frac{9}{20}$

d. $\frac{3}{20}$

9 In ΔABC, the circumcircle (O) has radius $R = 5$ and the incircle (I) has radius $r = 2$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 9

b. 15

c. 5

d. 6

10 In ΔABC, the circumcircle (O) has radius $R = 8$ and the incircle (I) has radius $r = 3$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 25

b. 16

c. 40

d. 17

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11 In ΔABC, the circumcircle (O) has radius $R = 11$ and the incircle (I) has radius $r = 4$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 33

b. 49

c. 77

d. 34

12 In ΔABC, the circumcircle (O) has radius $R = 14$ and the incircle (I) has radius $r = 5$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 81

b. 126

c. 57

d. 56

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